



A multi-agent collaborative framework with human-in-the-loop for well log interpretation and applications



CHEN Zhangxing^{1,2}, DING Ruichen^{2,3}, MENG Yang^{2,3}, LI Yizheng^{2,3}, ZHANG Junwei^{2,4}, CAO Liu^{2,5}, LI Jian², FAN Wenqi³, ZHANG Yiyuan³, WANG Liqiu³, ZHANG Dongxiao², CHEN Yuntian^{2,6,*}

1. Department of Chemical and Petroleum Engineering, University of Calgary, Calgary, AB T2N1N4, Canada;

2. Eastern Institute of Technology Ningbo Key Laboratory of Low-Carbon Hydrogen Energy, Ningbo 315200, China;

3. The Hong Kong Polytechnic University, Hong Kong 999077, China;

4. University of Science and Technology of China, Hefei 230026, China;

5. China University of Petroleum (Beijing), Beijing 102249, China;

6. Eastern Institute of Technology Zhejiang Key Laboratory of Industrial Intelligence and Digital Twin, Ningbo 315200, China

Abstract: This paper proposes a multi-agent system centered on large language models to address the issues that traditional well log interpretation relies on expert experience and poses great difficulty in multi-disciplinary collaboration and constructs a digital twin architecture across three dimensions of agents, tools and environment. At the agent level, a role-based architecture is established to decompose the complex log interpretation workflow into independent subtasks, enabling structured transfer of expert knowledge. At the tool level, petrophysical formulas and machine learning algorithms are encapsulated to form a physics-data dual-path hybrid reasoning mechanism; at the environment level, a standardized digital twin space is established based on the Model Context Protocol to achieve closed-loop control of the entire workflow. Engineers can drive the system through natural language commands to complete the full log interpretation process from data loading and parameter calculation to reservoir classification, realizing end-to-end automation from raw data to interpretation conclusions. In tests on 100 field wells, the system generates key interpretation parameters that are highly consistent with expert results, exhibiting stable recognition capability for complex reservoir types. This study demonstrates that this human-machine collaborative working mode significantly enhances the standardization and efficiency of well log interpretation, providing technical reference for intelligent transformation of highly specialized industrial processes.

Key words: complex reservoir types; well log interpretation; large language models; multi-agent system; digital twin space; hybrid reasoning; machine learning; human-machine collaboration

Introduction

Oil and gas exploration and development is critical to ensuring national energy security, involving multidisciplinary collaboration across geology, geophysical prospecting, reservoir engineering, and computer science [1–4]. Well logging interpretation bridges geological exploration and engineering decision-making, encompassing both standardized computational workflows and a strong reliance on expert empirical judgment, making it a typical scenario that exemplifies this mode of interdisciplinary collaboration [5–6]. In recent years, large language models (LLMs) have demonstrated emergent capabilities in

natural language understanding, cross-domain knowledge association, and complex logical reasoning [7–9], and have shown outstanding performance in tasks such as scientific discovery, code writing, literature summarization, and answering domain-specific questions [10–15]. These advancements provide a technical pathway for the intelligent automation of well logging interpretation.

Currently, the application of large language models in the geosciences and petroleum industry remains in its early stages, largely limited to leveraging their knowledge comprehension and generation capabilities for isolated or auxiliary tasks. In terms of knowledge management, re-

Received date: 13 Jan. 2026; **Revised date:** 15 May 2026.

* **Corresponding author.** E-mail: ychen@eitech.edu.cn

Foundation item: Supported by the Basic Science Center Project of National Natural Science Foundation of China (52288101); National Key Research and Development Program - Disruptive Technology Innovation Project (2024YFF1500600); National Natural Science Foundation Project of China (T2350710231).

[https://doi.org/10.1016/S1876-3804\(26\)60734-3](https://doi.org/10.1016/S1876-3804(26)60734-3)

Copyright © 2026, Research Institute of Petroleum Exploration and Development Co., Ltd., CNPC (RIPED). Publishing Services provided by Elsevier B.V. on behalf of KeAi Communications Co., Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

search has focused on constructing domain-specific pre-trained models (e.g., the geoscience model K2^[16]), as well as employing named entity recognition and knowledge graph technologies to enable information extraction and integration from unstructured geological reports and literature^[17–18], or applying them to geological terminology queries^[19] and core image segmentation^[20]. For data management, retrieval-augmented generation technology oriented toward subsurface domains has enabled automated parsing and intelligent retrieval of unstructured data such as production logs and vertical seismic profiles^[21]. In terms of engineering decision-making, multi-agent systems have begun to be applied in areas such as well log curve matching^[22], legacy well curve reconstruction and potential zone identification^[23], full-lifecycle downhole maintenance^[24], and autonomous optimization of wellbore operation strategies^[25]. However, current multi-agent architectures have yet to penetrate the core computational and reasoning workflows of well logging interpretation. Current software remains heavily dependent on expert experience for layer-by-layer parameter coordination and anomaly identification, lacking the capacity for continuous geological context reasoning and adaptive optimization, and thus struggling to meet the demands of high-frequency iterative engineering workflows.

To address the above issues, we developed a multi-agent collaborative system driven by large language models. At the agent level, the system is designed following the team structure and role division of field engineering teams, with the standard well logging interpretation workflow as the main thread for task decomposition and coordination. At the tool level, physical formulas for shale content, porosity, permeability, and other parameters are integrated with data-driven methods to construct a physics-data dual-driven hybrid reasoning pathway, balancing result interpretability and accuracy. At the environment level, existing well logging interpretation tools and data processing scripts are encapsulated as standard interfaces, allowing agents to invoke professional tools for computational tasks without modifying the existing production system. Engineers can intervene at key decision nodes via natural language, grounding the large model's task planning capability in concrete engineering computations. This work aims to address practical challenges in well logging interpretation, specifically the high dependence on expert experience and the difficulty of multi-professional collaboration. This is achieved by automatically coordinating professional tools, intelligently selecting parameters, and automatically reviewing results, thereby transitioning from tool-driven to intelligent decision-making and improving the efficiency and decision quality of complex reservoir evaluation.

Test results based on 100 field wells show that the system can independently complete the full well logging

interpretation workflow, with reservoir classification results broadly consistent with expert interpretations, and demonstrate the capability to extend from parameter computation to comprehensive evaluation. The system can dynamically optimize formula parameters based on geological understanding and real-time data, and generate traceable reasoning records, making full use of the contextual understanding capability of large language models. Through a role-based architecture and interpretable reasoning chains, the system structures expert experience and makes workflows transparent, providing a feasible pathway for oil and gas exploration to transition from experience-guided manual operation to knowledge-guided intelligent operation.

1. Multi-agent collaborative architecture for well log interpretation

In the system, each agent takes a large language model as its reasoning core and integrates perception, reasoning, action, and memory modules, thereby transforming the base language model into an agent system capable of autonomous decision-making and task execution in a controlled environment. The perception module is responsible for collecting and structurally compressing environmental information. The reasoning module decomposes task objectives into executable plans based on semantic understanding. The action module performs specific computations by invoking predefined tools. The memory module stores task history and domain knowledge, providing contextual support for continuous reasoning. The four modules are connected through standardized interfaces, forming a closed loop from environmental perception to action feedback, and ensuring safe and controllable execution under a permission-isolation mechanism, ultimately constructing an intelligent well logging interpretation system with both autonomy and reliability.

1.1. Core components of the agent

The perception module acquires information from the external environment and converts unstructured data into text summaries processable by the large language model. This module obtains the locations of data files, scripts, and reports through environment scanning tools, and retrieves statistical summaries and key features using data analysis tools. This design both reduces the impact of context length limitations of the large language model and aligns with expert analysis habits.

The reasoning module serves as the decision-making core of the agent, receiving environmental information and task instructions from the perception module, and generating structured action plans after analysis. These plans are defined in JSON (JavaScript Object Notation) format, explicitly specifying the tools to be invoked, the

parameters to be passed, and the execution sequence, thereby transforming engineer intent into instruction sequences that are machine-understandable and machine-executable.

The action module is responsible for translating the decisions of the reasoning module into actual operations, parsing instructions from the plan and invoking predefined tools to complete data processing and computation. This module constitutes the sole interface through which the agent interacts with the working environment, ensuring the safety and traceability of all operations.

The memory module stores both long-term and short-term memory. Long-term memory is preserved in the form of structured logs, forming complete and traceable task records. Short-term memory is derived from document retrieval augmentation and recent interaction history, including inter-agent communications, tool invocation records, and user instructions, providing the necessary contextual support for coherent task execution.

1.2. Hierarchical framework of the agent system

Fig. 1 illustrates the top-down four-layer architecture of the agent system: the human-AI interaction layer, the multi-agent collaboration layer, the individual execution layer, and the result feedback layer. The system supports

both continuous expert intervention and autonomous regulation. In experimental validation, the four-layer architecture is fully integrated across levels, forming an intelligent assistance system for the complete well logging interpretation workflow that achieves bidirectional collaboration between human intervention and autonomous regulation.

The human-AI interaction layer serves as the system control interface. The chief engineer agent is the core of this layer, receiving engineer instructions via natural language control. Engineers set task objectives, input geological understanding, and control process requirements, while the chief engineer agent is responsible for task planning and decomposition, generating standardized workflows, and issuing directives to the domain-expert agent engineering team.

The multi-agent collaboration layer is primarily responsible for task assignment and process control. Inspired by the well logging interpretation workflow, this layer constructs a domain-expert agent engineering team with clearly defined role divisions. The core collaboration logic is as follows: the chief engineer agent receives and analyzes engineer instructions, and converts them into task planning and allocation for domain expert agents. Domain expert agents are assigned to specific positions

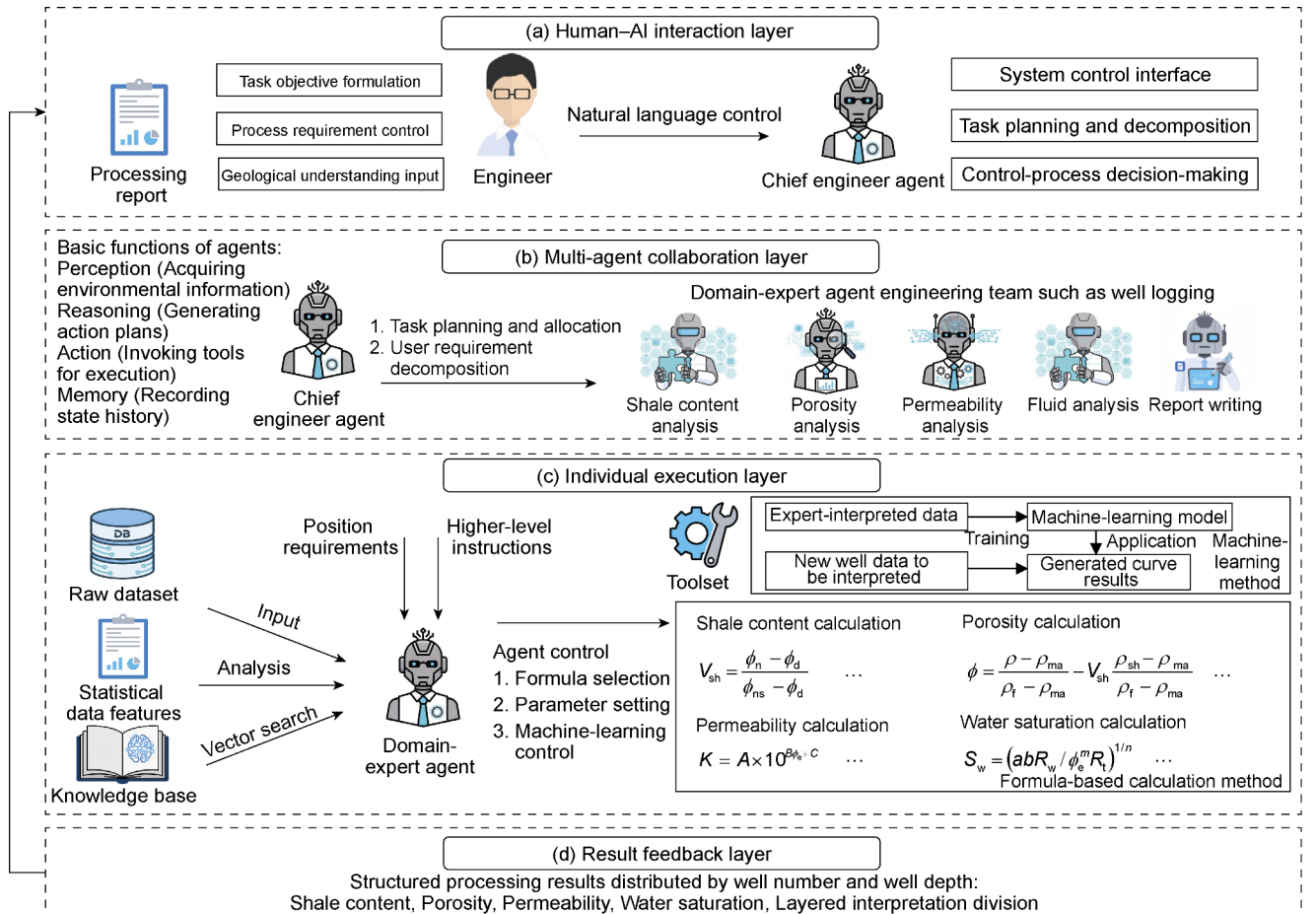


Fig. 1. Hierarchical framework design of agents for well log interpretation.

following the well logging interpretation workflow, such as shale content analysis, porosity analysis, permeability analysis, water saturation analysis. In the workflow, each domain expert agent corresponds to a specific toolset, thereby reducing the operational complexity of individual roles and improving success rates.

Domain expert agents can access five categories of inputs: access permissions to the raw dataset, statistical data features, knowledge base information, standard position requirements, and higher-level instructions transmitted by the chief engineer agent, including processing requirements and geological understanding that engineers wish to convey. Agents compute interpretation parameters through either formula-based calculation methods or machine learning methods. In the formula-based calculation method, the agent system invokes petrophysical formulas and sets empirical parameters according to the geological context. The machine learning module integrates several mainstream algorithms, including Extreme Gradient Boosting (XGBoost), Multilayer Perceptron (MLP), Long Short-Term Memory (LSTM), and Transformer. The system dynamically constructs input–output combinations based on data completeness and task requirements, with well logging curves (e.g. gamma ray, interval transit time, and resistivity) as inputs and target parameters (e.g. porosity and permeability) as outputs. Under machine-learning control, agents analyze available data, automatically select appropriate curve combinations as inputs, and determine output parameters based on computational objectives. In practice, multiple optional formulas may exist for the same interpretation parameter, and certain key parameters within the formulas require adjustment according to the block-specific geological context. Domain expert agents are typically required to complete three categories of intelligent decisions: formula selection, empirical parameter setting, and machine-learning control for configuring input–output curves.

The result feedback layer integrates interpretation results, including shale content, porosity, permeability, water saturation and reservoir division conclusions. These results are presented in structured data format for convenient engineer review. Based on the computed results and actual requirements, engineers can submit revision requests to the system through the human-AI interaction layer, forming a closed loop from result feedback to optimization and adjustment.

1.3. Agent collaboration methods

In industrial engineering scenarios oriented toward stable production, ensuring the reliability of task planning is paramount. To this end, a quantitative analysis of the planning complexity of agent collaboration modes is conducted to identify their scalability bottlenecks and to

demonstrate the necessity of introducing a structured collaboration architecture.

To characterize the scale of tool combination search by agents under unconstrained conditions, the planning complexity can be modeled as:

$$C_{\text{free}} = \sum_{i=1}^{L_{\text{max}}} N^i \quad (1)$$

Eq. (1) demonstrates that the combination of increasing tool quantity and lengthening task chains triggers a combinatorial explosion in the global search space. Upon introducing the agent collaboration framework, the planning complexity under a multi-agent collaborative architecture can be expressed as:

$$C_{\text{arch}} = M \sum_{i=1}^{K_i} \sum_{l=1}^{L_{i\text{max}}} S_i^l \quad (2)$$

In Eq. (2), M characterizes the additional overhead incurred among multiple agents for coordinating task execution. Its value directly determines the collaboration efficiency and overall scalability of the system, making it the core optimization target of this study. Upon completion of a staged task, the system must designate the next executor among multiple agents. In the absence of a standardized assignment mechanism, the open selection space leads to repeated invocation of the same agent and redundant communication, causing M to grow exponentially with the number of task stages. In contrast, by drawing on mature standard operating procedures from the industrial domain and implementing clearly defined role divisions and step optimization, M can be constrained within a small constant upper bound. In this case, the planning complexity of the agent divide–conquer architecture is primarily limited by the longest tool chain. Since the toolset undergoes specialized decomposition, the conditions $S_i < N$ and $L_{i\text{max}} \leq L_{\text{max}}$ are satisfied. Therefore, this collaborative architecture can effectively constrain the scale of the planning space and enhance the reliability of system execution.

The above analysis demonstrates that by introducing standardized process constraints and functional decomposition, the originally exponentially growing complexity planning problem can be decomposed into controllable sub-problems, thereby improving the scalability of the system.

1.4. A multi-agent collaboration system inspired by the well logging interpretation workflow

The structured multi-agent collaboration mode is grounded in the standard well logging interpretation workflow, mapping real-world responsibility divisions onto the agent system. As illustrated in Fig. 2, the system defines an agent organizational structure and communication protocols, forming a virtual expert team centered on the chief engineer agent as the scheduling hub, with multiple domain expert agents operating collaboratively. Each domain expert agent is equipped with a dedicated

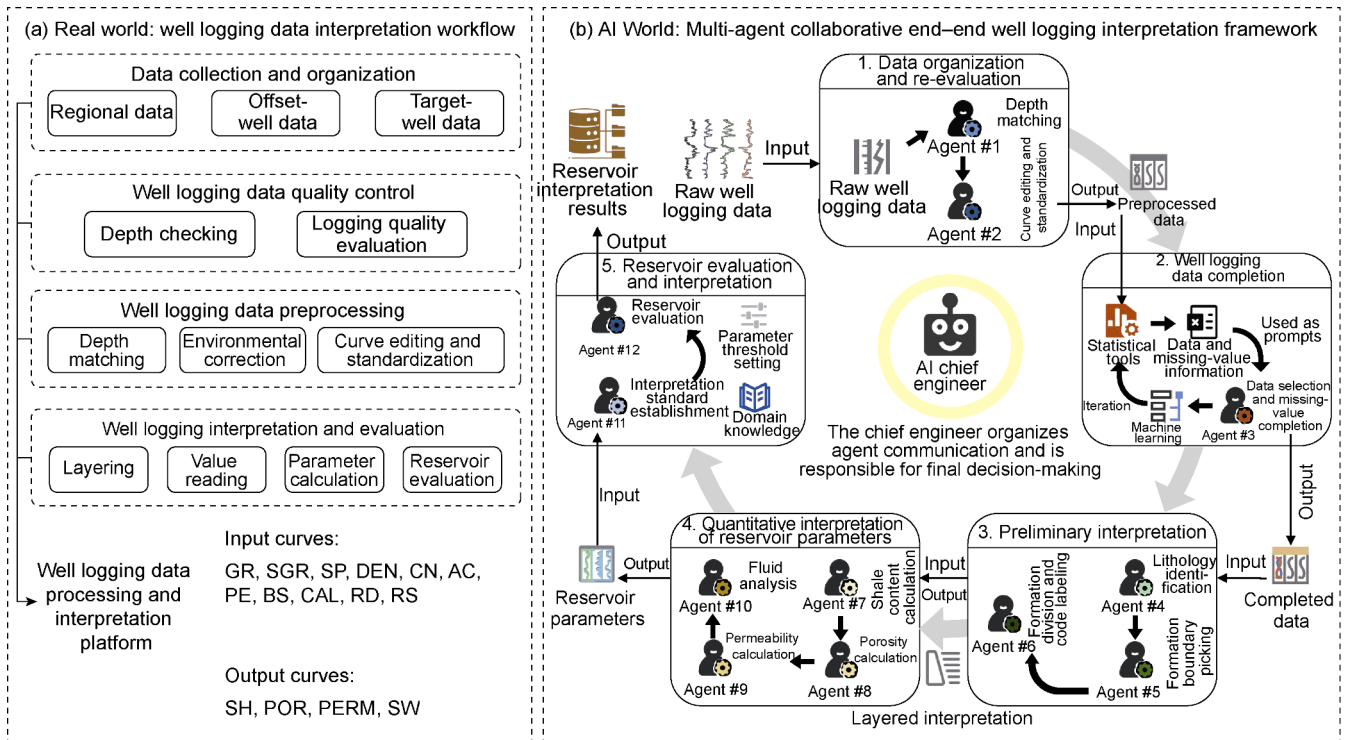


Fig. 2. Multi-agent collaborative system inspired by the well logging analysis and interpretation workflow.

toolset and independently completes specialized tasks assigned by the chief engineer agent under a position responsibility system, achieving the unification of workflow standardization and execution specialization.

In real-world well logging data interpretation, the complete workflow encompasses data collection and organization, well logging data quality inspection, well logging data preprocessing, and well logging interpretation and evaluation, each stage governed by clearly defined technical specifications. Following field operation procedures, this design establishes corresponding domain expert agents for specific tasks including well logging data completion, shale content calculation, porosity calculation, permeability calculation, and fluid analysis. Each agent has clearly defined responsibilities and is equipped with a dedicated toolkit. The core advantage of this design lies in mapping real-world workflows and responsibility divisions, validated through long-term practice, onto the agent organization to form a structured collaboration mechanism. This approach lowers the threshold for system design and reduces trial-error costs, while institutionally embedding domain knowledge within the system architecture.

The communication mechanism of the agent system adopts the Model Context Protocol (MCP). This protocol is an open-source interface standard for AI applications, designed to standardize interactions between systems. Based on MCP, inter-agent collaboration and external tool invocation can be realized within a modular and decoupled framework. Agent interactions are conducted via MCP and recorded as structured logs by the memory

module, forming auditable execution traces and shared context that ensure consistency and transparency throughout complex collaborative processes.

2. Operational logic of the multi-agent system

Fig. 3 illustrates the operational logic of the agents. Upon program initialization, a virtual team comprising multiple domain expert agents is instantiated, with each agent's role, responsibilities, and capability scope defined via prompts, and dedicated knowledge bases loaded according to task requirements. During the execution phase, agents operate through a closed-loop "observation-action-feedback" cycle: acquiring state information from the environment, invoking tools to complete well logging interpretation steps, and adjusting subsequent actions based on feedback. The environment encompasses three primary elements: raw well logging data and intermediate results; an agent interaction architecture grounded in standard well logging interpretation workflows; and a toolkit developed for well logging interpretation, covering both theoretical formula computation and machine learning inference modules. The tool layer serves as the engineered extension of agent capabilities and constitutes the sole outlet for numerical outputs. Agents drive tool execution through control files, retaining the authority to observe computational results but not to directly modify numerical values. This mechanism ensures that all outputs are processed through verifiable physical or mathematical tools, thereby suppressing the influence of model hallucinations on processing results.

Fig. 4 illustrates the operational logic of the system.

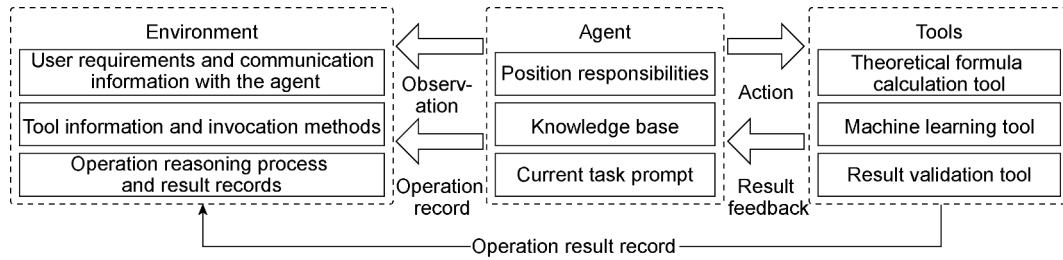


Fig. 3. Single-agent design.

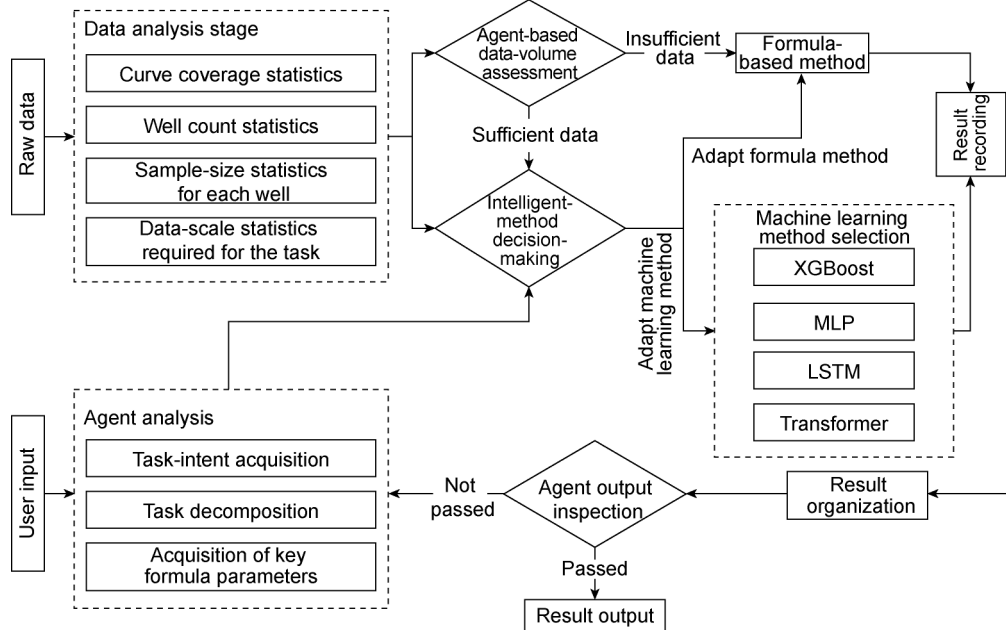


Fig. 4. Multi-agent operational logic relationship and system decision-making process.

After the engineer sets objectives in natural language, the chief engineer agent receives the instructions, parses the task intent, and decomposes the workflow, specifying the domain expert agents to be invoked and their collaboration sequence. The agents invoke function tools to analyze the curve coverage, well count, and sample size of the raw logging data required for the task. Concurrently, geological prior knowledge extracted from user inputs is written into configuration files. The system automatically performs a data sufficiency assessment. If the available data fails to meet the training requirements of machine learning models, physics-based formula computation tools are activated; if the data is sufficient, the agent decision module dynamically selects between formula-based methods and machine learning algorithms. Computational results undergo verification for physical consistency and statistical rationality; results that fail verification are routed back to the analysis stage for iterative reasoning. The successful results will be output along with the explanation, which will be used to update the environment log. This workflow design ensures full-chain automation and controllability across data diagnosis, path selection, and result verification.

3. Experimental design and results

To empirically evaluate the system's capability in han-

dling complex engineering problems, this study designed a two-phase experimental framework: (1) comparing the effectiveness of tool-chain planning across different agent collaboration modes; and (2) conducting end-end automated well logging interpretation tasks to quantitatively assess the comprehensive performance of the proposed architecture in real-world engineering scenarios. The evaluation focused on three primary dimensions: human-AI collaboration capability, well logging interpretation accuracy, and system operational efficiency. Five computational experiments were conducted, in which the error characteristics of system outputs were analyzed by varying human-AI interaction strategies and data partitioning schemes, to assess the stability and applicable scope of the system.

3.1. Agent collaboration modes

Through controlled experiments, the performance of three agent collaboration modes in tool-chain planning tasks was compared. The three tested operational modes exhibit distinct differences in architectural design and permission configuration. The single-agent tool-chain mode relies on a single agent to independently complete the entire tool-chain planning. The multi-agent negotiation mode deploys three agents that generate the tool chain through two rounds of planning discussion and

Table 1. Comparison of multiple agent collaboration modes

Statistical metric	Pass count rate/%	Input characters	Output characters	Planning runtime/s	Number of test cases
Single-agent tool-chain mode	18	411 805	45 311	303.16	100
Multi-agent negotiation mode	27	875 115	219 689	1 927.39	100
Agent domain specialization mode	82	532 079	89 784	1 055.72	100

voting-based decision-making. The agent domain specialization mode constructs domain expert agents, with each agent strictly constrained to a designated tool subset and professional stage, collaboratively generating the tool chain. To evaluate the planning accuracy and execution efficiency of different agent collaboration modes, a standardized task repository comprising 100 test cases was constructed. Each case was designed based on real-world well logging interpretation scenarios, consisting of manually authored task prompts and target tool chains. The target tool chain defines the correct sequence of tool invocations required to complete the given task. The tool library used in the experiments simulates the core procedures of practical well logging interpretation, encompassing 19 specialized tools, including key modules such as data loaders, curve correction, and well logging interpretation parameter computation. Based on the task prompts, agents autonomously plan the complete tool execution path. A result is counted as a success only when the tool chain generated by the agent is entirely consistent with the predefined target.

The experimental results in Table 1 confirm the performance differences among different agent collaboration modes in tool-chain planning tasks. In the multi-agent negotiation mode, all agents have access to the full tool-set and lack structured collaboration constraints, causing the planning search space and coordination costs to expand, resulting in the longest runtime (1 927.39 s), the highest communication overhead (875 115 input characters), and a task pass rate of only 27%. The single-agent mode has the shortest runtime (303.16 s) and the lowest communication volume, but the full-tool search leads to high planning difficulty, with a success rate of only 18%. The agent domain specialization mode, through rational partitioning of tool subsets and standardization of collaborative workflows, reduces planning complexity, and outperforms the multi-agent negotiation mode in pass rate (82%), runtime (1 055.72 s), and communication cost.

3.2. Well logging interpretation task evaluation

3.2.1. Experimental dataset and evaluation criteria

The well logging interpretation results are empirically evaluated using an industrial-grade dataset comprising 100 wells of field logging data. Each well contains multiple types of sensor response curves, including lithology, physical properties, and electrical properties. The core task of the agent system is to transform raw logging data into two categories of geological information: obtaining

key petrophysical interpretation parameters (shale content, porosity, permeability, and water saturation); and performing qualitative reservoir classification.

In actual operations, well logging data frequently suffer from missing data problems due to cost control and varying requirements at different exploration stages. In addition, missing values and outliers caused by physical measurement limitations or interference during data acquisition also occur within individual curves. This leads to an exponential increase in the combinatorial complexity of interpretation tasks. Traditional approaches rely on expert empirical judgment, requiring substantial human effort. The agent-based well logging interpretation system, by contrast, needs to assess the current state of the data and autonomously search for valid interpretation paths under the constraints of domain knowledge and optimization objectives, achieving the transition from experience-driven to systematized intelligent reasoning.

The data-driven models within the system are trained on a reserved subset of well data and calibrated using a small amount of high-precision core analysis data. During the evaluation phase, the system takes as input well data that were not involved in training and completes the full workflow from data diagnosis to final interpretation. The experiments use expert field results as the reference, employing accuracy to evaluate the computational precision of key interpretation parameters, and comprehensively assessing reservoir classification conclusions using accuracy, recall, and F1 score.

3.2.2. Human-AI interaction analysis

In the agent system testing, engineers participate in the full workflow of well logging data processing by inputting operational instructions and geological knowledge in natural language. The large language model encodes natural language instructions into token sequences for processing. Fig. 5 presents statistics on token consumption during system operation.

The content generated by the agents covers the full workflow from data preprocessing to result output, including task decomposition instructions, tool invocation parameters, sub-task coordination information, and process optimization suggestions. Fig. 5 shows that the characters input by engineers account for only 1.43% of all control characters, indicating that the agent team assumes the primary control responsibilities, while engineers shift toward a management role focused on objective setting and critical decision-making. This

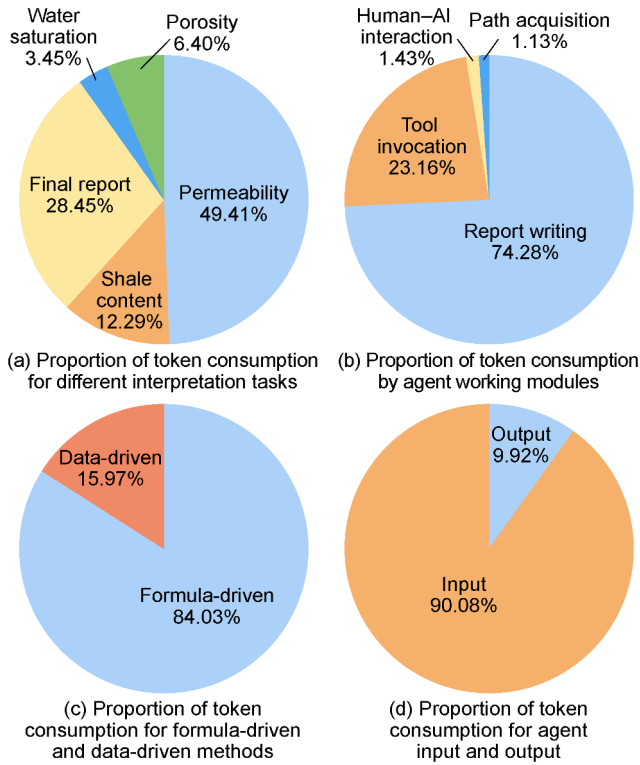


Fig. 5. Statistical chart of token consumption proportions during system operation.

human-AI collaboration mode effectively reduces operational workload while ensuring the professionalism of well logging processing. The token consumption proportions for different interpretation parameters show that permeability calculation requires more reasoning characters, indicating greater reasoning demands. Formula-driven methods consume more characters than data-driven methods, suggesting that they require more geological context and data analysis to support formula selection and parameter adjustment.

Fig. 6 illustrates the process of human-AI collaboration optimization of shale content calculation in the system. Sandstone matrix density and formation water density are key parameters in formula-based computation. In principle, both should be determined based on data from the entire block; however, due to the limitations of local well logging data, corrections incorporating the engineer's block-level geological knowledge are required. The chief engineer agent parses the geological prior knowledge input by the user and transmits it to the shale content calculation agent. The latter performs method reasoning based on local well logging data characteristics and role constraints, autonomously determines the computation strategy, generates configuration files, and executes the process step by step according to stratigraphic age, outputting formatted results for engineer review. The experimental results demonstrate that by introducing geological prior knowledge (adjusting the prompt) in place of direct instructions, the mean absolute error (MAE) of shale content calculation decreased from

47.08% to 16.60%, demonstrating the effectiveness of human-AI collaboration.

This process demonstrates that, when processing specific formations, the system is capable of dynamically selecting formulas and determining key parameters based on geological knowledge and data analysis, while providing clear reasoning justifications, thereby achieving context-aware adaptive optimization. By structuring expert experience, the system ensures computational reliability while acquiring the capabilities of dynamic judgment and logical reasoning.

3.2.3. Calculation performance of interpretation parameters

The agent system integrates physics-based formula computation and data-driven machine learning algorithms, forming a hybrid reasoning framework with complementary strengths. The theoretical formula method is grounded in petrophysical theory and well logging engineering formulas, offering clear physical meaning and strong interpretability, and is capable of providing robust geological inferences even in new blocks lacking prior data. The data-driven module integrates various machine learning methods, including XGBoost, MLP, LSTM, and Transformers. It leverages their feature extraction and nonlinear fitting capabilities to mine complex mapping relationships between well logging responses and reservoir parameters derived from historical data. Additionally, it accurately learns the localized empirical interpretation embedded in engineers' processing workflows for specific blocks.

This paper uses expert interpretations determined by engineers based on comprehensive geological understanding, petrophysical analysis, and multi-well comparisons as a benchmark to quantitatively evaluate the consistency of the calculation results of the intelligent agent system. To eliminate the influence of differences in the dimensions and value ranges of different interpretation parameters on the evaluation results, the accuracy based on the range normalization error is used as an evaluation index.

In the statistical evaluation of porosity and permeability calculation accuracy, to eliminate the interference of lithological mixture on the statistical results, the analysis scope is restricted to non-mudstone intervals, and five repeated experiments are conducted to obtain the range of result fluctuations, thereby enhancing the statistical effect of the evaluation. Table 2 presents the accuracy results of the agent system in calculating well logging interpretation parameters. The results show that the machine learning models achieve accuracy exceeding 90% in calculations of all the four reservoir parameters.

This indicates that machine learning models are capable of accurately fitting the processing results and decision

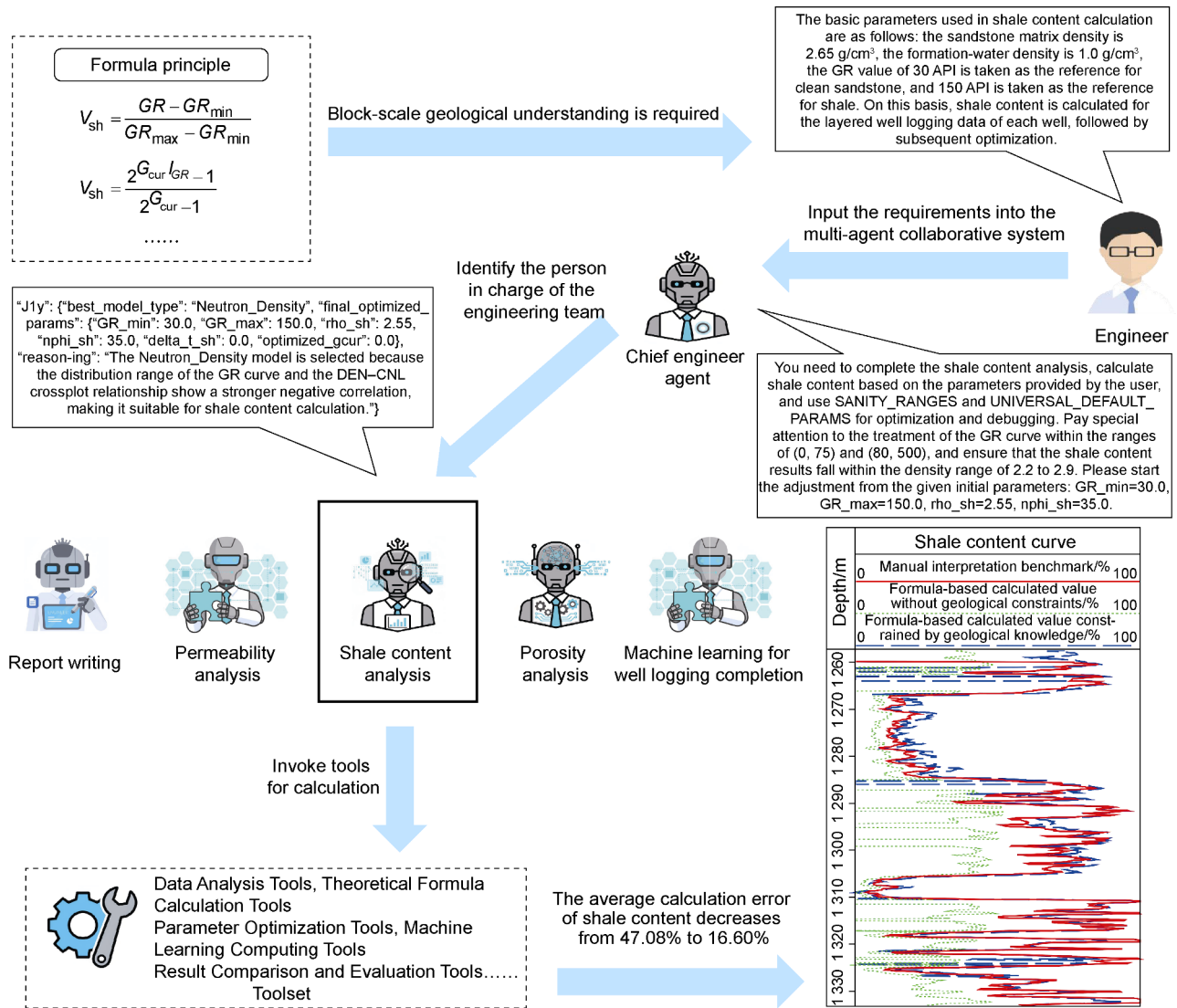


Fig. 6. Human-in-the-loop process control illustration.

Table 2. Accuracy in calculating well logging interpretation parameters

Well logging interpretation Parameter	Accuracy of formula-based method/%	Accuracy in different machine learning models/%			
		XGBoost	MLP	LSTM	Transformer
Shale content	82.14±1.77	92.16±0.51	91.17±0.47	91.04±0.49	92.05±0.01
Porosity	73.11±6.30	95.41±0.49	90.91±0.76	92.23±0.56	95.90±0.53
Permeability	96.92±0.70	98.73±0.88	98.78±0.78	98.62±0.77	98.57±0.61
Water saturation	75.70±0.01	94.83±0.26	93.64±0.22	94.25±0.28	94.64±0.32

patterns of field engineers. In practical well logging interpretation, when core data are limited, engineers typically select or construct empirical formulas based on the geological characteristics of the study area — a process that essentially constitutes a standardized treatment of geological relationships. The well logging interpretation agent system precisely captures this parameter mapping relationship through machine learning algorithms, thereby achieving a high degree of consistency with engineer interpretation results. However, this high accuracy also has its limitations. The black-box nature of machine learning models makes their decision-making process difficult to reconstruct into an intelligible reasoning chain, and engineers find it hard to trace the origin of the results. Model parameters are obtained through mathe-

matical optimization and bear no direct semantic correspondence to geological variables such as porosity and saturation, resulting in insufficient physical interpretability and making them unreliable as a basis for geological decision-making.

The formula method, as a classical approach to well logging interpretation, constructs analytical expressions based on physical principles or geological experience. This method offers clear physical meaning and strong interpretability, allowing engineers to understand the formula parameters and make adjustments in conjunction with geological knowledge. However, in practical applications, due to the complexity and heterogeneity of geological conditions, prior geological knowledge is often incomplete, and the key parameters of formulas fre-

quently rely on limited core data or empirical estimation. This simplification based on limited information often leads to systematic deviations between the calculated results and the actual reservoir characteristics. As shown in Table 2, the formula method yields lower calculation accuracy than machine learning methods across all interpretation parameters. It is worth noting that, due to the wide distribution range of permeability values in field well logging data, range-normalization-based accuracy is susceptible to interference from a small number of extreme values during evaluation, which may introduce bias into the assessment results. To more robustly compare the computational performance of different methods, Table 3 lists the mean absolute error as a supplementary reference.

The agent system integrates well logging theoretical formulas and machine learning algorithms as computational modules, forming a physics-data dual-path hybrid reasoning mechanism that can be dynamically selected and invoked according to task requirements. The theoretical formula module, grounded in physical principles,

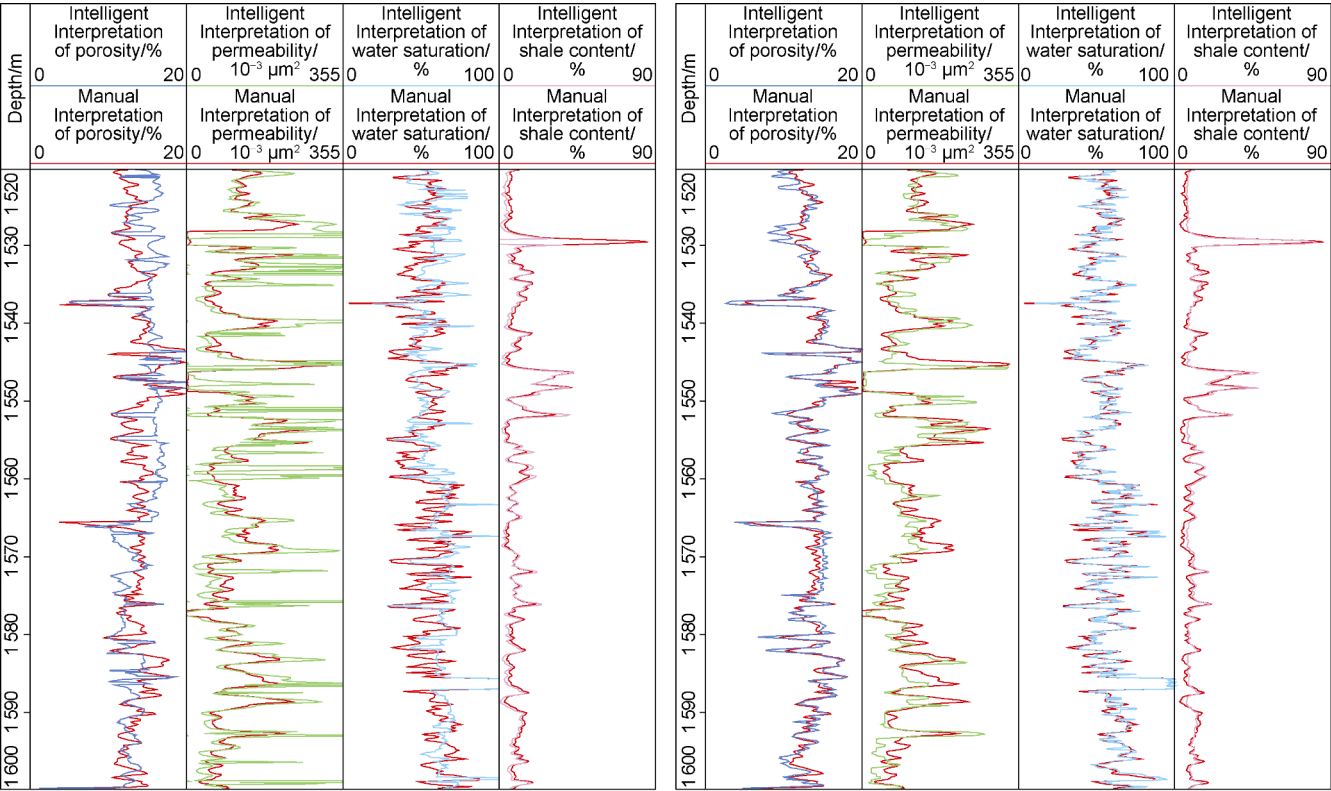
provides an interpretable reasoning chain. The machine learning module achieves high-precision computation by learning the expert processing patterns of the study area.

Fig. 7 compares the results of the theoretical formula method, the machine learning method, and field engineer interpretations over the same well interval. For the shale content parameter, both computational methods show good agreement with the manual interpretation results. However, in the calculation of porosity, permeability, and saturation parameters, the results of the theoretical formula method exhibit deviations from the engineer interpretations (Fig. 7a).

This discrepancy stems from differences in working methodology. In practical processing, engineers typically use core calibration data to establish proprietary empirical formulas tailored to the geological characteristics of specific blocks, whereas the theoretical formula library invoked by the agents primarily consists of general well logging formulas and petrophysical models. The key parameters of general formula systems are difficult to determine directly through theoretical derivation, as they

Table 3. Mean absolute error of the agent system in calculating well logging interpretation parameters

Well logging interpretation parameter	Mean absolute error			
	Shale content/%	Porosity/%	Permeability/($10^{-3} \mu\text{m}^2$)	Water saturation/%
Formula-based method	18.09±1.79	8.87±2.08	14.23±8.88	21.64±0.01
XGBoost	7.75±0.52	1.34±0.04	4.26±0.85	4.58±0.34
MLP	8.74±0.49	2.76±0.08	3.98±0.73	5.73±0.33
LSTM	8.87±0.51	2.33±0.12	3.94±0.83	5.17±0.38
Transformer	8.00±0.42	1.24±0.10	4.24±0.85	4.81±0.39



(a) Comparison between theoretical-formula-based interpretation results and manual interpretation results

(b) Comparison between machine-learning-based interpretation results and manual interpretation results

Fig. 7. Comparison of the effects of intelligent interpretation parameters and manual interpretation parameters.

are susceptible to the influences of reservoir heterogeneity, microstructure, and fluid properties. Without targeted calibration, fixed parameters are unable to adapt to the response mechanisms of specific geological bodies, resulting in systematic offsets in the calculated results. In contrast, machine learning methods, by virtue of their powerful nonlinear fitting capabilities, can directly learn complex mapping relationships between inputs and outputs from data, thereby achieving a high degree of consistency with manual interpretation results. Although this data-driven modeling approach demonstrates superior predictive performance, its decision-making process is difficult to translate into physically interpretable or verifiable mechanisms and empirical rules that geologists can understand and validate.

Fig. 8 presents the shale content calculation results, where the horizontal and vertical axes represent the engineer-interpreted values and the agent system-calculated values of shale content, respectively, and the red diagonal line represents perfect agreement between the calculated results and the engineer's processed results, with greater deviation from this line indicating poorer consistency. The comparison between the theoretical formula method results and the manual interpretation results in the figure exhibits systematic errors, reflecting the constraints imposed by the preset physical model framework and the limitations in empirical parameter knowledge, as well as insufficient adaptability in intervals with stronger formation heterogeneity, and such deviations are a manifestation of the formula-based approach's inherent characteristics. The data points of the machine learning method are closely distributed along the diagonal line, demonstrating a high degree of reproducibility of the engineer's processing patterns. Its data-driven mechanism is capable

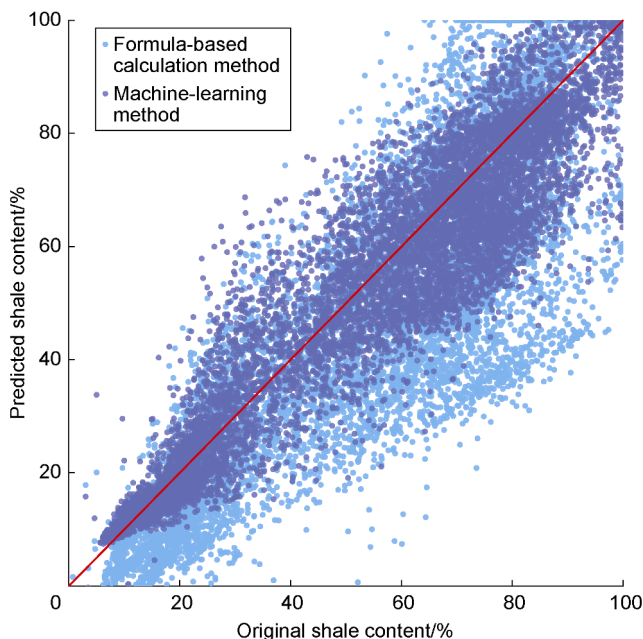


Fig. 8. Shale content calculation results.

Table 4. Time requirements of formula-based methods for key well logging parameters

Well logging interpretation parameter	Workflow time/s	Computation time/s
Shale content	110.64±27.60	14.89±17.97
Porosity	136.96±30.53	29.32±19.34
Permeability	696.51±367.95	38.31±17.72
Water saturation	314.72±493.47	51.33±14.24

of implicitly learning nonlinear mapping relationships under complex geological conditions, exhibiting stronger data-adaptive characteristics, and providing a high-precision technical pathway for well logging interpretation.

Table 4 presents statistics on the computation runtime of the calculation methods for four key well logging parameters, namely shale content, porosity, permeability, and water saturation, within the agent system, covering two metrics: overall workflow runtime and core computation runtime. In the traditional mode, completing full-parameter interpretation for 100 wells typically requires 1 to 2 weeks, and relies on manual well-by-well parameter adjustment and result validation, with efficiency subject to the influence of personnel experience. In contrast, the maximum runtime for single-parameter calculation in this system is less than 20 min, significantly reducing the time cost of well logging interpretation and making it suitable for rapid batch processing of well logging data, thereby improving operational efficiency. The computation runtime shown in the table is far less than the total workflow runtime, indicating that the agents' inferential decision-making has become the primary bottleneck of the workflow. The computation runtime for permeability and water saturation exhibits large fluctuations, stemming from the need to perform inference for each combination of formation and rock types, where the combinatorial explosion causes a sharp increase in the frequency of large model invocations, and the fluctuations of individual invocations are cumulatively amplified.

3.2.4. Evaluation on stratification results

Reservoir qualitative classification is the final stage of this system's workflow, and its performance serves as an important basis for evaluating the end-end interpretation results of the system. In this stage, the machine learning engineer agent is responsible for constructing a classification model using the XGBoost algorithm to handle high-dimensional well logging features. To address the class imbalance problem where economically valuable but sample-scarce categories such as oil layers are underrepresented, the synthetic minority over-sampling technique (SMOTE) is integrated into the workflow to balance the data distribution, reduce the risk of model bias toward the majority class, and enhance the identification capability for key reservoir types.

Table 5 and Table 6 present the model performance

Table 5. Manual interpretation performance of reservoir classification prediction on raw data

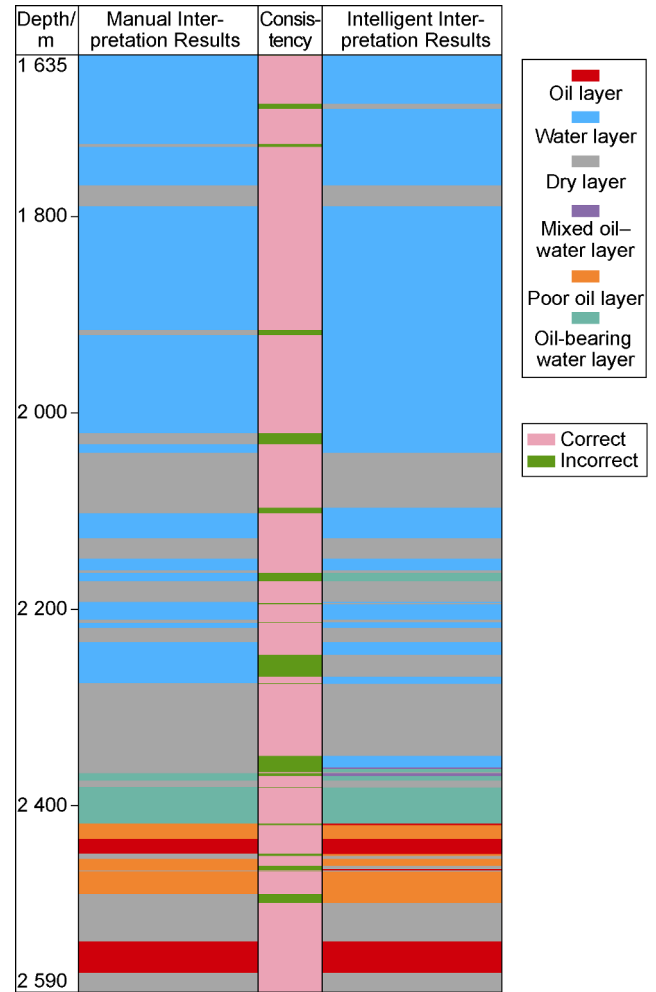
Reservoir type	Accuracy/ %	Recall/ %	F_1 /%	Sample size
Oil-bearing water layer	40	72	51	2 249
Poor oil layer	47	70	56	2 332
Dry layer	85	80	82	13 444
Water layer	96	86	90	33 929
Oil layer	68	74	71	2 125
Mixed oil-water layer	33	82	47	909
Weighted average	87	82	84	54 988

Table 6. Interpretation performance of reservoir classification prediction with intelligent interpretation incorporated

Reservoir type	Accuracy/ %	Recall/ %	F_1 /%	Sample size
Oil-bearing water layer	40	79	53	2 249
Poor oil layer	51	73	60	2 332
Dry layer	86	81	84	13 444
Water layer	97	88	92	33 929
Oil layer	75	77	76	2 125
Mixed oil-water layer	52	89	66	909
Weighted average	88	85	86	54 988

using raw well logging data (gamma ray, neutron porosity, spontaneous potential, formation resistivity, density, and acoustic transit time) and after incorporating intelligent interpretation parameters (shale content, permeability, porosity, and water saturation). The evaluation is based on the interpretation results of 20 independent wells. The overall weighted score (F_1) improved from 84% to 86%. The limited improvement is primarily attributable to data distribution: water layers and dry layers together account for more than 86% of the samples, and their baseline performance is already relatively high ($F_1 > 80\%$), constraining the room for improvement in overall metrics and highlighting the progress made in critical minority classes. However, in small-sample and high-value categories, the model achieves important breakthroughs: F_1 for mixed oil-water layers increases from 47% to 66%, representing an improvement of 40.4%; that for layers improve from 71% to 76%, and that for poor oil layers from 56% to 60%, both exceeding 7% improvement. These categories, which collectively account for less than 10% of the total samples, correspond to key reservoir types that are difficult to discriminate yet carry high industrial value, and the enhancement of their identification capability holds practical application value for exploration decision-making. The results demonstrate that the introduction of intelligent interpretation parameters enhances the model's discriminative capability for complex reservoirs, with particularly notable improvement in the identification of mixed oil-water layers.

Fig. 9 presents the reservoir classification prediction results through a depth profile comparison. The left side of the figure shows the actual reservoir, while the right

**Fig. 9. Validation of reservoir classification prediction performance for hydrocarbon-bearing intervals in an exploration well.**

side shows the predicted reservoir, with “Correct” (pink zone) indicating correctly predicted intervals and “Incorrect” (green zone) indicating incorrectly predicted intervals. Overall, the predicted results show strong consistency with the actual reservoir in terms of vertical distribution. In intervals where oil layers are concentrated, correct predictions (pink zone) predominate, demonstrating that the model achieves stable and reliable oil layer identification. In intervals characterized by variable fluid responses, the model maintains a higher prediction consistency, reflecting its adaptability to diverse lithologies and physical properties. Across the entire profile, the vertically continuous nature of the predictions effectively captures the vertical distribution patterns of reservoirs, offering a valuable reference for vertical reservoir characterization.

4. Conclusions

Following the digital mapping logic from the real world to the twin world, the intelligent well logging interpretation system is constructed across three dimensions, namely agent roles, professional tools, and operational

environment, forming an engineering agent-based paradigm that is practically deployable.

The agent role design establishes a position-based architecture grounded in large language models, decomposing global planning and professional analysis responsibilities among the chief engineer agent and domain expert agents for shale content, porosity, and other parameters. By explicitly defining capability boundaries and specializing toolsets, the complexity of long planning is reduced. This hierarchical collaboration, which simulates real-world expert teams, achieves the structured inheritance of expert experience through the explicit binding of position responsibilities and knowledge bases.

The professional tool layer transforms well logging domain knowledge into callable modules by encapsulating a library of petrophysical theoretical formulas, thereby enhancing engineering interpretability. Meanwhile, leveraging the reasoning capabilities of large language models, petrophysical theoretical formulas and machine learning algorithms are jointly encapsulated as structured operational modules, establishing a physics-data dual-path decision mechanism. This enables the system to automatically switch to physical formulas when data are scarce to ensure system stability, and to invoke machine learning models when data are sufficient to approach expert-level processing, achieving adaptive decision-making under complex data conditions. In addition, the system adopts an isolation design between configuration-driven and tool-execution layers, prohibiting large models from directly generating numerical results, thereby suppressing the risk of model hallucination.

The operational environment establishes a standardized runtime space based on the Model Context Protocol (MCP). Drawing on standard well logging interpretation workflows, this environment establishes an operational procedure of data collection, data quality inspection, data preprocessing, and interpretation evaluation, ensuring that multiple agents effectively leverage industrial practice experience for task orchestration and intermediate result handoff. The digital twin environment relies on structured log analysis to form traceable decision records, enabling full-process controllability from data input to interpretation conclusions.

Experiments conducted on 100 wells demonstrate that the system is capable of autonomously constructing and executing complete well logging interpretation workflows, with final interpretation results highly consistent with expert conclusions. This outcome validates the potential of multi-agent collaborative architectures in addressing real-world engineering problems.

Although the intelligent well logging interpretation system has demonstrated its effectiveness, the current implementation still has several limitations. Regarding tools, although the system integrates existing well logging

interpretation tools and provides standardized extension interfaces, its operation is strictly confined to a predefined toolset. As large models evolve toward embodied intelligence, agents will need the capability to continuously learn from the environment and solve new problems. However, the constraints of the predefined toolset prevent agents from independently writing or debugging new scripts to handle unexpected issues. Therefore, developing secure sandbox environments for agents to autonomously write, debug, and execute codes would enhance their flexibility. Regarding agents, this work validates the effectiveness of the divide-conquer architecture, which ensures system stability by drawing on organizational experience and incorporating standard well logging workflows. However, the agent interaction structure still relies on manual presets, limiting its adaptive adjustment capability when confronted with new scenarios. Future work should explore structure optimization mechanisms based on environmental feedback to achieve autonomous optimization of agent collaboration structures. Regarding the environment, although the system constructs a twin environment that aggregates data and logs, the influx of information as the scale of information and the number of tools grow will lead to increased processing noise. How to optimize agent memory mechanisms and interaction efficiency while ensuring context integrity, and suppress interference from irrelevant information, is an important research direction for improving system scalability.

In summary, this system constructs a practically deployable human-AI collaborative solution for well logging interpretation through the synergistic design of agent roles, professional tools, and operational environment across three levels. Despite current limitations in tool autonomous evolution, agent dynamic topology, and environmental information denoising, the framework has validated its engineering feasibility in large-scale testing across 100 wells. By replacing the domain knowledge base and expert agents, the architecture can be extended to other geoscience problems such as seismic interpretation and reservoir numerical simulation, providing a reference for the intelligent upgrading of highly specialized industrial workflows. This work validates the core value of human-AI collaboration: experts are responsible for top-level geological logic and critical decision supervision based on intuition and experience, while AI systems undertake knowledge integration and computational execution through standardized protocols, with the two complementing each other. As the capabilities of underlying models continue to improve, agent systems of this kind, characterized by human-in-the-loop supervision and structured design as the core, are expected to play an increasingly important role in a broader range of complex engineering tasks.

Nomenclature

A, B, C, a, b —empirical coefficients;
 C_{arch} —planning complexity under multi-agent collaborative architecture;
 C_{free} —planning complexity without structural constraints;
 G_{CUR} —stratigraphic age correction factor;
 GR —gamma ray, API;
 GR_{min} —minimum gamma ray value of pure sandstone, API;
 GR_{max} —maximum gamma ray value of pure mudstone, API;
 K —permeability, $10^{-3} \mu\text{m}^2$;
 K_i —number of task stages;
 l —tool chain length set during planning;
 L_{max} —upper limit of reasonable planning steps;
 $L_{i\text{max}}$ —maximum planning step length of subtasks undertaken by the agent at the i -th task stage;
 M —agent collaboration complexity factor;
 m —cementation exponent;
 N —number of tools;
 n —saturation exponent;
 I_{GR} —gamma ray index;
 R_i —formation resistivity, $\Omega \cdot \text{m}$;
 R_w —formation water resistivity, $\Omega \cdot \text{m}$;
 S_i —size of the tool subset available to the agent at the i -th task stage;
 S_w —water saturation, %;
 V_{sh} —shale content, %;
 ϕ —total porosity, %;
 ϕ_e —effective porosity, %;
 ϕ_D —porosity calculated from density log, %;
 ϕ_N —neutron porosity, %;
 ϕ_{Ns} —neutron porosity of mudstone, %;
 ρ —formation bulk density, g/cm^3 ;
 ρ_f —pore fluid density, g/cm^3 ;
 ρ_{ma} —rock matrix density, g/cm^3 ;
 ρ_{sh} —mudstone density, g/cm^3 .

References

- [1] JIA Chengzao, PANG Xiongqi, SONG Yan. Basic principles of the whole petroleum system. *Petroleum Exploration and Development*, 2024, 51(4): 679–691.
- [2] WANG Xuejun, CAI Jiaming, WEI Xiaodong. The current status and development trend of geophysical technology for oil and gas exploration. *China Petroleum Exploration*, 2014, 19(4): 30–42.
- [3] LIU He, REN Yili, LI Xin, et al. Research status and application of artificial intelligence large models in the oil and gas industry. *Petroleum Exploration and Development*, 2024, 51(4): 910–923.
- [4] CHEN Zhangxing, ZHANG Yongan, LI Jian, et al. Artificial intelligence large model for logging curve reconstruction. *Petroleum Exploration and Development*, 2025, 52(3): 744–756.
- [5] LI Ning, XU Binsen, WU Hongliang, et al. Application status and prospects of artificial intelligence in well logging and formation evaluation. *Acta Petrolei Sinica*, 2021, 42(4): 508–522.
- [6] LI Ning, LIU Yingming, WANG Caizhi, et al. Construction and application of CIFLog logging digital-intelligent cloud platform in Daqing Oilfield. *Petroleum Geology & Oilfield Development in Daqing*, 2024, 43(3): 17–25.
- [7] OpenAI. GPT-4 technical report. (2023-03-15)[2024-12-05]. <https://arxiv.org/abs/2303.08774>.
- [8] Gemini Team, Google. Gemini: A family of highly capable multimodal models. (2023-12-07)[2024-12-05]. <https://arxiv.org/abs/2312.11805>.
- [9] Anthropic. The Claude 3 model family: Opus, Sonnet, Haiku. [2024-12-05]. https://www-cdn.anthropic.com/de8ba9b01c9ab7cbabf5c33b80b7bbc618857627/Model_Card_Claude_3.pdf.
- [10] BOIKO D A, MACKNIGHT R, GOMES G. Emergent autonomous scientific research capabilities of large language models. (2023-04-11)[2024-12-05]. <https://arxiv.org/abs/2304.05332>.
- [11] XU F F, ALON U, NEUBIG G, et al. A systematic evaluation of large language models of code: CHAUDHURI S, SUTTON C. *Proceedings of the 6th ACM SIGPLAN International Symposium on Machine Programming*. New York: Association for Computing Machinery, 2022: 1–10.
- [12] SUSNJAK T, HWANG P, REYES N, et al. Automating research synthesis with domain-specific large language model fine-tuning. *ACM Transactions on Knowledge Discovery from Data*, 2025, 19(3): 68.
- [13] TSCHISGALE P, MAUS H, KIESER F, et al. Evaluating GPT- and reasoning-based large language models on Physics Olympiad problems: Surpassing human performance and implications for educational assessment. *Physical Review Physics Education Research*, 2025, 21(2): 020115.
- [14] JIANG X, WANG W R, TIAN S H, et al. Applications of natural language processing and large language models in materials discovery. *npj Computational Materials*, 2025, 11(1): 79.
- [15] PAGANO S, STRUMOLO L, MICHALK K, et al. Evaluating ChatGPT, Gemini and other Large Language Models (LLMs) in orthopaedic diagnostics: A prospective clinical study. *Computational and Structural Biotechnology Journal*, 2025, 28: 9–15.
- [16] DENG C, ZHANG T H, HE Z M, et al. K2: A foundation language model for geoscience knowledge understanding and utilization: ANGÉLICA L, LATTANZI S, MEDINA A M, et al. *Proceedings of the 17th ACM International Conference on Web Search and Data Mining*. New York: Association for Computing Machinery, 2024: 161–170.
- [17] CHEN S Y, HUA W H, LIU X G, et al. Chinese fine-grained geological named entity recognition with rules and FLAT. *Earth and Space Science*, 2022, 9(12): e2022EA002617.
- [18] TANG X M, FENG Z Q, XIAO Y T, et al. Construction and

- application of an ontology-based domain-specific knowledge graph for petroleum exploration and development. *Geoscience Frontiers*, 2023, 14(5): 101426.
- [19] ECKROTH J, GIPSON M, BODEN J, et al. Answering natural language questions with OpenAI's GPT in the petroleum industry. SPE 214888-MS, 2023.
- [20] REN Yili, ZENG Changmin, LI Xin, et al. Intelligent evaluation of sandstone rock structure based on a visual large model. *Petroleum Exploration and Development*, 2025, 52(2): 488–498.
- [21] KUMAR P, MAZROUEI B S A, FAYYAZ M F, et al. Realizing the need for agentic AI for subsurface data and workflows. SPE 229586-MS, 2025.
- [22] XIONG Wenjun, XIAO Lizhi, YUAN Jiangru, et al. Automatic depth matching method of well log based on deep reinforcement learning. *Petroleum Exploration and Development*, 2024, 51(3): 553–564.
- [23] CHEN H, ZHAO Q, LU W, et al. Application of self-supervised autonomous agent framework with growing scheme for digital transformation of elder/existed well potentials discovery. IPTC 24776-MS, 2025.
- [24] CHEN Y N, GU S J, ZHANG M, et al. An LLM-based multi-agent system for whole life-cycle downhole maintenance operations. SPE 229606-MS, 2025.
- [25] CHEN H, JIN W, WANG Y, et al. Swarm AIGC agent based approach for shale oil wellbore operation policy decision supporting: Field implementation in western China. SPE 229645-MS, 2025.